OSPF Router ID

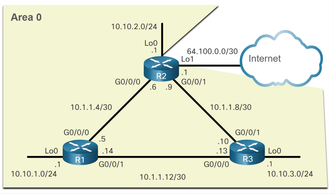
2.1.1

OSPF Reference Topology

To get you started, this topic discusses the foundation on which OSPF bases its entire process, the OSPF router ID.

The figure shows the topology used for configuring OSPFv2 in this module. The routers in the topology have a starting configuration, including interface addresses. There is currently no static routing or dynamic routing configured on any of the routers. All interfaces on R1, R2, and R3 (except the loopback 1 on R2) are within the OSPF backbone area. The ISP router is used as the gateway to the internet of the routing domain.

Note: In this topology the loopback interface is used to simulate the WAN link to the Internet and a LAN connected to each router. This is done to allow this topology to be duplicated for demonstration purposes on routers that only have two Gigabit Ethernet interfaces.



2.1.2

Router Configuration Mode for OSPF

OSPFv2 is enabled using the **router ospf** *process-id* global configuration mode command, as shown in the command window for R1. The *process-id* value represents a number between 1 and 65,535 and is selected by the network administrator. The *process-id* value is locally significant, which means that it does not have to be the same value on the other OSPF routers to establish adjacencies with those neighbors. It is considered best practice to use the same *process-id* on all OSPF routers.

After entering the **router ospf** *process-id* command, the router enters router configuration mode, as indicated by the **R1(config-router)#** prompt. Enter a question mark (?), to view all the commands available in this mode. The list of commands shown here has been altered to display only the commands that are relevant to this module.

```
R1(config)# router ospf 10
R1(config-router)# ?
  area            OSPF area parameters
  auto-cost       Calculate OSPF interface cost according to bandwidth
  default-information  Control distribution of default information
  distance        Define an administrative distance
  exit            Exit from routing protocol configuration mode
  log adjacency-changes  Log changes in adjacency state
  neighbor        Specify a neighbor router
  network         Enable routing on an IP network
  no              Negate a command or set its defaults
  passive-interface  Suppress routing updates on an interface
  redistribute    Redistribute information from another routing protocol
  router-id       router-id for this OSPF process
R1(config-router)#
```

2.1.3

Router IDs

An OSPF router ID is a 32-bit value, represented as an IPv4 address. The router ID is used to uniquely identify an OSPF router. All OSPF packets include the router ID of the originating router. Every router requires a router ID to participate in an OSPF domain. The router ID can be defined by an administrator or automatically assigned by the router. The router ID is used by an OSPF-enabled router to do the following:

- **Participate in the synchronization of OSPF databases** – During the Exchange State, the router with the highest router ID will send their database description (DBD) packets first.
- **Participate in the election of the designated router (DR)** – In a multiaccess LAN environment, the router with the highest router ID is elected the DR. The routing device with the second highest router ID is elected the backup designated router (BDR).

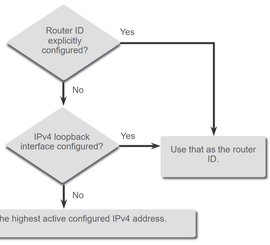
Note: The DR and BDR election process is discussed in more detail later in this module.

2.1.4

Router ID Order of Precedence

But how does the router determine the router ID? As illustrated in the figure, Cisco routers derive the router ID based on one of three criteria, in the following preferential order:

1. The router ID is explicitly configured using the OSPF **router-id** *rid* router configuration mode command. The *rid* value is any 32-bit value expressed as an IPv4 address. This is the recommended method to assign a router ID.
2. If the router ID is not explicitly configured, the router chooses the highest IPv4 address of any of configured loopback interfaces. This is the next best alternative to assigning a router ID.
3. If no loopback interfaces are configured, then the router chooses the highest active IPv4 address of any of its physical interfaces. This is the least recommended method because it makes it more difficult for administrators to distinguish between specific routers.



2.1.5

Configure a Loopback Interface as the Router ID

In the reference topology, only the physical interfaces are configured and active. The loopback interfaces have not been configured. When OSPF routing is enabled on the router, the routers would pick the following highest active configured IPv4 address as the router ID:

- R1: 10.1.1.14 (G0/0/1)
- R2: 10.1.1.9 (G0/0/1)
- R3: 10.1.1.13 (G0/0/0)

Note: OSPF does not need to be enabled on an interface for that interface to be chosen as the router ID.

Instead of relying on physical interface, the router ID can be assigned to a loopback interface. Typically, the IPv4 address for this type of loopback interface should be configured using a 32-bit subnet mask (255.255.255.255). This effectively creates a host route. A 32-bit host route would not get advertised as a route to other OSPF routers.

The example shows how to configure a loopback interface on R1. Assuming the router ID was not explicitly configured or previously learned, R1 will use IPv4 address 1.1.1.1 as the router ID. Assume R1 has not yet learned a router ID.

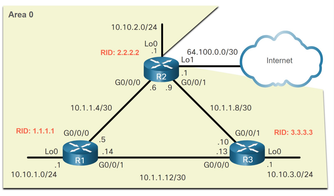
```
R1(config-if)# interface Loopback 1
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# end
R1# show ip protocols | include Router ID
  Router ID 1.1.1.1
R1#
```

2.1.6

Explicitly Configure a Router ID

In the figure, the topology has been updated to show the router ID for each router:

- R1 uses router ID 1.1.1.1
- R2 uses router ID 2.2.2.2
- R3 uses router ID 3.3.3.3



Use the **router-id** *rid* router configuration mode command to manually assign a router ID. In the example, the router ID 1.1.1.1 is assigned to R1. Use the **show ip protocols** command to verify the router ID.

```
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
R1(config-router)# end
*May 23 15:31:42.680: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip protocols | include Router ID
  Router ID 1.1.1.1
R1#
```

2.1.7

Modify a Router ID

After a router selects a router ID, an active OSPF router does not allow the router ID to be changed until the router is reloaded or the OSPF process is reset.

In example for R1, the configured router ID has been removed and the router reloaded. Notice that the current router ID is 10.10.1.1, which is the Loopback 0 IPv4 address. The router ID should be 1.1.1.1. Therefore, R1 is configured with the command **router-id 1.1.1.1**.

Notice how an informational message appears stating that the OSPF process must be cleared or that the router must be reloaded. The reason is because R1 already has adjacencies with other neighbors using the router ID 10.10.1.1. These adjacencies must be renegotiated using the new router ID 1.1.1.1. Use the **clear ip ospf process** command to reset the adjacencies. You can then verify that R1 is using the new router ID command with the **show ip protocols** command piped to display only the router ID section.

Clearing the OSPF process is the preferred method to reset the router ID.

```
R1# show ip protocols | include Router ID
  Router ID 10.10.1.1
R1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
% OSPF: Reload or use "clear ip ospf process" command, for this to take effect
R1(config-router)# end
R1# clear ip ospf process
Reset ALL OSPF processes? [no]: y
*Jun  6 01:09:46.979: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/1 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun  6 01:09:46.979: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun  6 01:09:46.981: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done
*Jun  6 01:09:46.981: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
R1# show ip protocols | include Router ID
  Router ID 1.1.1.1
R1#
```

Note: The **router-id** command is the preferred method. Otherwise, the router chooses the highest IPv4 loopback interface address or the highest active IPv4 address of any of its physical interfaces.

2.1.8

Syntax Checker - Configure R2 and R3 Router IDs

Use the Syntax Checker to configure R2 and R3 with router IDs.

```
You are currently logged into R2:

  - Enter OSPF router configuration mode using process ID 10
  - Assign a router ID of 2.2.2.2
  - Return to privileged EXEC mode


R2(config)# router ospf 10
R2(config-router)#
```

Reset | Show Me | Show All

2.1.9

Check Your Understanding - OSPF Router ID

Check your understanding of OSPF router ID by choosing the BEST answer to the following questions.

1. True or False? In the **router ospf** *process-id* command, the process ID value, which can any number between 1 and 65,535, is locally significant. It must be the same on all routers in the OSPF area.

   ○ True
   ◉ False

2. Which of the following applies to the router ID? (Choose two)

   ☐ The router ID cannot be defined by an administrator.
   ☐ The router ID is not used to determine the BDR.
   ☑ The router ID is used to determine the DR.
   ☑ The router ID uniquely identifies the router.
   ☐ The router ID is not required.

3. Which of the following is the order of precedence for choosing the router ID?

   ○ 1. Router ID that is explicitly configured
     2. Lowest IPv4 loopback address
     3. Lowest active configured IPv4 address

   ○ 1. Router ID that is explicitly configured
     2. Lowest active configured IPv4 address
     3. Lowest IPv4 loopback address

   ◉ 1. Router ID that is explicitly configured
     2. Highest IPv4 loopback address
     3. Highest active configured IPv4 address

   ○ 1. Highest IPv4 loopback address
     2. Router ID that is explicitly configured
     3. Highest active configured IPv4 address

   ○ 1. Highest IPv4 loopback address
     2. Highest active configured IPv4 address
     3. Router ID that is explicitly configured

Check | Show Me | Reset